

GenAI Powered Data Analytics

End-to-End Delinquency Risk Analytics & Agentic AI Collections Framework

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SECTION

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Executive Summary

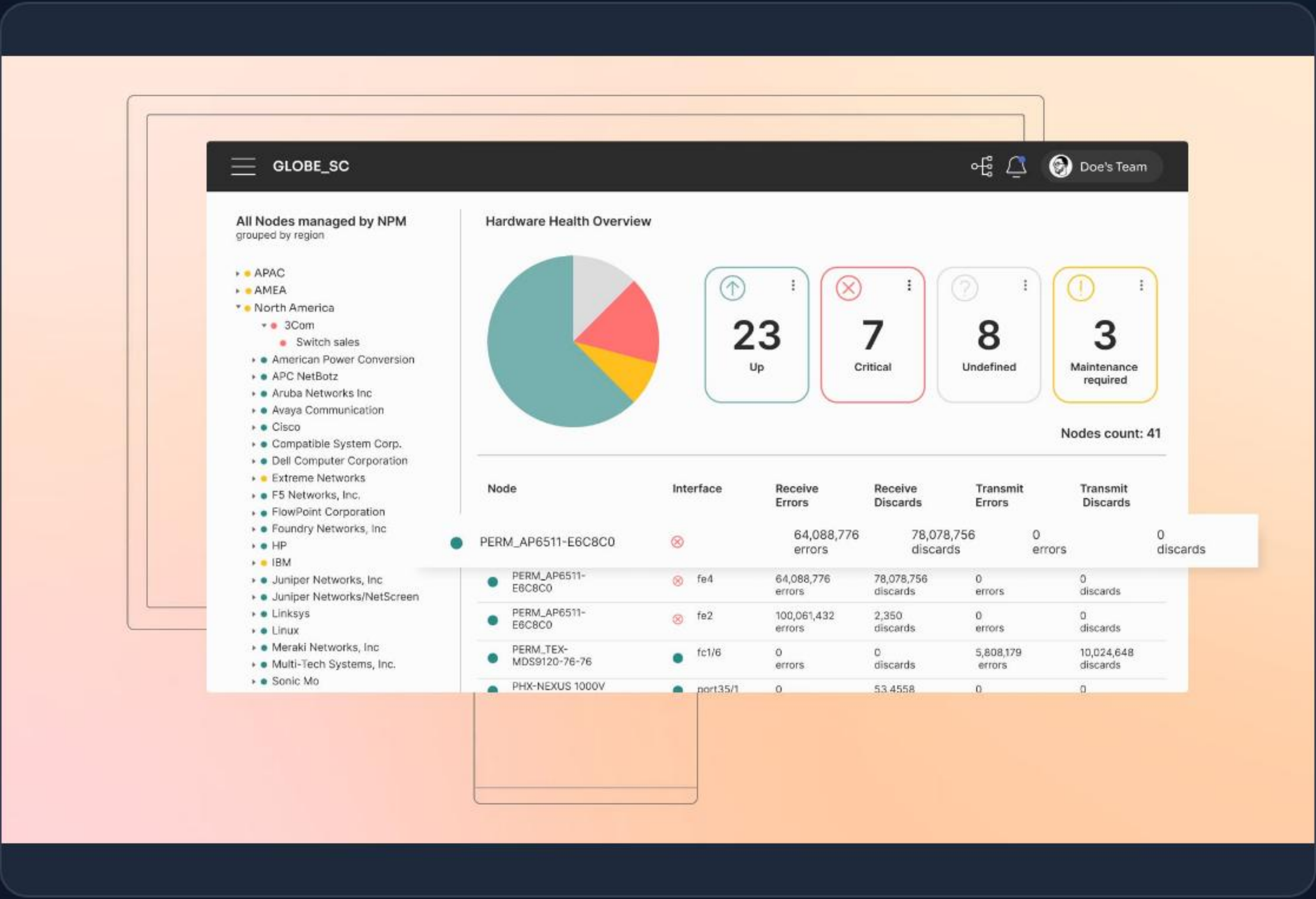
A comprehensive AI transformation framework designed for Geldium Finance to predict credit card delinquency and deploy ethical, automated collections strategies.

Project Scenario & Role Brief

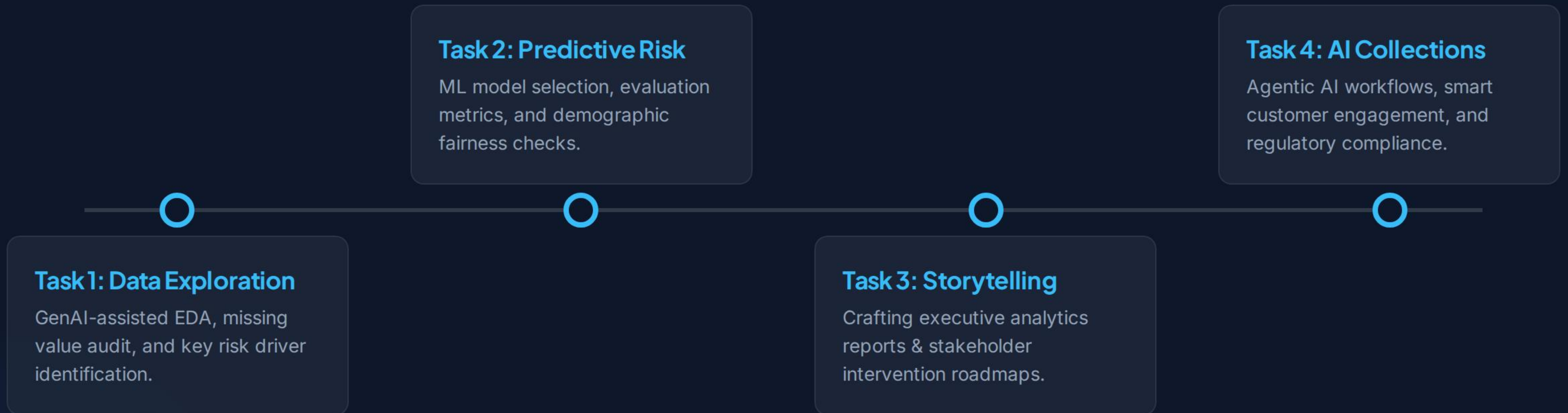
Role: AI Transformation Consultant

Targeted problem: Geldium Finance faced rising credit card delinquency rates. The collections team needed advanced analytics to identify high-risk customers early and optimize outreach.

Objective: Leverage Generative AI alongside classic predictive ML to perform exploratory data analysis, select optimal risk models, and engineer an automated, fair collections strategy.



| Job Simulation Workflow



| Task 1: EDA & Risk Profiling

GenAI-Assisted Data Audit

Automated prompt workflows identified data imbalance (18.4% default rate) and flagged missing value patterns across credit score and utilization metrics, structuring clean feature inputs.

Core Risk Indicators

Key indicators strongly correlated with delinquency included high credit limit utilization (>75%), repeated late payment history (30+ days past due), and recent multi-card inquiries.

| Task 2: Model Performance



The XGBoost model, combined with GenAI-engineered behavioral features, yielded superior discriminatory power (0.91 ROC-AUC) for detecting early customer credit default.

| Ethical & Responsible AI



Fairness Auditing

Enforced equalized odds across age, gender, and income demographics to eliminate algorithmic bias in credit risk scoring.



SHAP Explainability

Utilized SHAP value calculations to provide clear human-understandable reason codes for every automated risk flag.



Human-in-the-Loop

High-stakes actions, such as credit limit reductions or legal escalations, require mandatory human compliance review.

| Task 3: Portfolio Risk Target

24.5%

Projected Delinquency Reduction

Executive Leadership Insights

By shifting from reactive collection calls to proactive GenAI-assisted interventions at Day 15 of delinquency, Geldium Finance can mitigate \$12.4M in potential bad-debt write-offs annually.

| Task 4: Agentic AI Outreach

Smart Collections Engine

Deploys conversational GenAI agents to reach out to at-risk clients via preferred channels (WhatsApp, SMS, Email) with dynamic payment restructuring options based on customer hardship signals.

Projected Delinquency Rate



Delinquency rate forecast demonstrates a sharp decline following the full deployment of GenAI automated risk scoring and smart collection workflows in 2024.

| Simulation Key Takeaways



GenAI-Driven EDA: Leveraged prompt engineering to accelerate exploratory analysis, missing value detection, and risk profiling.



Predictive Risk Modeling: Evaluated ML frameworks (XGBoost, Random Forest) to optimize credit delinquency classification accuracy.



Responsible AI Frameworks: Implemented bias checks, demographic parity audits, and regulatory compliance standards for financial automation.



Strategic Business Communication: Transformed quantitative model outputs into clear executive summaries and actionable collections recommendations.

Thank You!

GenAI Powered Data Analytics Job Simulation Complete

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| Image Sources



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